Emergent Spacetime and the Dark Sector

Probing Dark Matter via the ER = EPR *Conjecture in Holographic Tensor Networks*

Master of Science Thesis Prospectus • 40-Week Research Horizon (60 ECTS)

Abstract

The missing dynamical mass inferred from galactic kinematics remains an outstanding challenge to the standard ΛCDM paradigm, where decades of particle dark matter searches have yielded null bounds. This thesis develops a discrete, computational realization of emergent gravity, testing whether the phenomenon of dark matter can be accounted for by non-local quantum entanglement defects without particle relics. Grounded in the ER = EPR conjecture [5], Verlinde's entropic gravity [2], and fundamentally in the *Fractal Entanglement Quantum Gravity* (FEQG) framework formulated by Giné [1], continuous 2D spatial manifolds are discretized into hyperbolic tensor networks. Edge lengths follow Giné's information-theoretic metric $d_{ij} = -l_0 \ln(I_{ij}/I_0)$ [1], while non-local Planckian Einstein–Rosen bridges induce an informational stress tensor $E_{\mu\nu} \sim \nabla_\mu \nabla_\nu S_{\text{ent}}$ and modified potentials $V(r) \sim 1/r^{1+\epsilon}$ [1]. Utilizing graph-theoretic min-cut algorithms (discrete Ryu–Takayanagi [4]), we extract emergent radial potentials $\Phi(r)$ and circular velocity fields $v_c(r)$. The model is directly tested against 1D rotation curves from the SPARC database [6], 2D kinematic fields from SDSS/MaNGA, and gravitational microlensing constraints from the Large Magellanic Cloud (LMC/MACHO) [3].

1 Scientific Context and Original Contribution

1.1 Literature Status vs. The Scope of this Thesis

Recent analytical breakthroughs have challenged the particle dark matter paradigm by deriving gravitational backreaction from quantum information:

- **Verlinde (2016):** Postulated that matter displaces the volume-law thermal entropy of de Sitter space, producing an elastic reaction force that mimics dark matter below the critical acceleration $\Sigma < a_0/(8\pi G)$ [2].
- **Giné (2026):** Formulated the primary theoretical foundation of this work, *Fractal Entanglement Quantum Gravity* (FEQG) [1], deriving the master field equation $G_{\mu\nu} = 8\pi G(T_{\mu\nu} + \alpha E_{\mu\nu} + \beta F_{\mu\nu})$ from an underlying entanglement graph with the relational distance law $d_{ij} = -l_0 \ln(I_{ij}/I_0)$ [1].

The Unaddressed Research Gap: Both frameworks are strictly analytical, continuum-based, and lack numerical implementation on discrete geometries. In particular, Giné explicitly identifies the absence of numerical and computational tools as an open frontier in FEQG [1]. No prior work has constructed a discrete tensor network to simulate minimal-cut surfaces with non-local ER bridges and fitted the resulting velocity fields directly to empirical astrophysical data. **This thesis provides that computational execution and observational test.**

2 Theoretical Framework

In holographic tensor networks (e.g., MERA, HaPPY code), the entanglement entropy of boundary subregion A is given by the discrete Ryu–Takayanagi (RT) cut [4]:

$$S(A) = \min_{\gamma_A} \sum_{e \in \gamma_A} \ln(\chi_e) = \frac{\text{Area}(\gamma_A)}{4G\hbar}, \tag{1}$$

where χ_e is the bond dimension of cuts crossing the extremal surface γ_A.

To represent the dark sector, we superimpose an ensemble of gauge-neutral, non-local entanglement bonds (Planckian ER bridges) onto a 2D Poincaré disk discretization with connectivity distribution $\rho_{\text{ER}}(r)$. According to Giné's FEQG framework [1], edge weights follow mutual information decay:

$$d_{ij} = -l_0 \ln\left(\frac{I_{ij}}{I_0}\right). \tag{2}$$

Linearized perturbations around the vacuum satisfy $\delta S_{\text{ent}} = \delta\langle H_{\text{mod}}\rangle$. In the weak-field limit, the informational stress tensor $E_{\mu\nu} = \nabla_\mu \nabla_\nu S_{\text{ent}} - g_{\mu\nu}\Box S_{\text{ent}}$ introduced by Giné [1] modifies the metric component $h_{00} = -2\Phi_{\text{eff}}/c^2$, generating the effective orbital velocity:

$$v_c^2(r) = r\frac{\partial \Phi_{\text{eff}}}{\partial r} = v_{\text{bary}}^2(r) + v_{\text{ER}}^2(r). \tag{3}$$

The thesis systematically investigates whether the graph-theoretic min-cut on this perturbed network replicates the asymptotically flat rotation curves ($v_c \to$ const) observed in galaxies.

3 Computational Pipeline Architecture

The simulation engine is implemented as a modular Python pipeline:

1. **Graph Topology Generation (`NetworkX`, `igraph`):** Discretizes the 2D spatial slice into a hyperbolic tessellation with coordinates (r, θ).
2. **Baryonic Seeding:** Injects central vertex saturation and mass operators matching the visible gas and stellar disk profiles from SPARC ($M_{\text{gas}} + M_{\text{disk}}$) [6].
3. **ER Defect Superposition:** Seeds non-local chordal edges across distant coordinates with probability $P(r_i, r_j) \propto |r_i - r_j|^{-\gamma}$ and assigns weights via Eq. (2) [1].
4. **Min-Cut Solver (`scipy.sparse.csgraph`):** Solves the max-flow/min-cut duality via the Push-Relabel algorithm to determine $\gamma_A(r)$ and $\delta S_{\text{ent}}(r)$ for concentric boundary intervals.
5. **Differentiation & Kinematic Fitting (`SciPy`, `Lmfit`):** Computes radial derivatives to obtain $g(r) = -\nabla\Phi_{\text{eff}}$, extracts $v_c(r)$, and performs χ^2 minimizations against real galaxy profiles.

4 Empirical Observational Datasets

The model is evaluated against three distinct astrophysical datasets:

- **SPARC Database (175 Galaxies) [6]:** 1D high-precision rotation curves. Targets include classic spirals (e.g., NGC 3198, NGC 2841) and gas-rich, dark-matter-dominated Low Surface Brightness (LSB) dwarf galaxies.
- **SDSS-IV MaNGA:** 2D integral field spectroscopy velocity fields to evaluate whether the 2D tensor network preserves observed kinematic axisymmetry and shear.
- **LMC/MACHO Microlensing (Hubble/VLT) [3]:** Verifies that the model's non-local geometric deflection does not violate empirical bounds on compact halo objects (MACHO Collaboration).

5 40-Week Milestone Schedule

Phase	Timeline	Core Methodological Deliverable
Phase 1	Weeks 1–10	**Analytical Modeling:** Formal derivation of the discrete min-cut response to non-local link ensembles; proof of the $w \approx 0$ dust equation of state from Giné's tensor $E_{\mu\nu}$ [1].
Phase 2	Weeks 11–22	**Algorithm Construction:** Building and validating the hyperbolic graph generator and Push-Relabel min-cut engine; reproducing Keplerian baseline ($v_c \propto r^{-1/2}$).
Phase 3	Weeks 23–32	**Simulation & SPARC Ingestion:** Seeding ER links; executing multi-parameter sweeps; fitting simulated curves to SPARC 1D [6] and MaNGA 2D profiles.
Phase 4	Weeks 33–40	**Synthesis & Defense:** Computing comparative residual statistics (χ^2_{model} vs. χ^2_{NFW} vs. χ^2_{MOND}); compiling final thesis manuscript.

6 Seven-Chapter Master Dissertation Syllabus

1. **Introduction and the Galactic Missing Mass Crisis:** Observational foundations of flat rotation curves; Tully–Fisher relation; failure of Newtonian baryonic models; standard ΛCDM halo profiles (NFW, Burkert); the status of particle dark matter direct detection.
2. **Quantum Information Geometry and ER = EPR:** Spacetime from entanglement; the Maldacena–Susskind conjecture [5]; Planckian Einstein–Rosen bridges; topological absence of Standard Model gauge couplings; tensor networks as toy cosmologies.
3. **Holographic and Fractal Entanglement Thermodynamics:** Ryu–Takayanagi formula [4]; Jacobson–Faulkner variational field equations; Verlinde's volume-law entropic displacement [2]; Giné's FEQG framework and dimensional flow ($D \to 2$) [1].
4. **The Entanglement Defect Model of Dark Matter:** Analytical formulation of non-local ER defect ensembles; derivation of the informational stress tensor $E_{\mu\nu}$ [1]; weak-field metric potential $\Phi_{\text{eff}}(r)$; demonstration of effective dust-like behavior.
5. **Discrete Numerical Engine and Algorithmic Architecture:** Mapping 2D hyperbolic spatial disks to discrete graphs; edge-weighting via $d_{ij} = -l_0 \ln(I_{ij}/I_0)$ [1]; max-flow/min-cut graph algorithms; central baryonic mass injection.

6. **Simulation Results and Astrophysical Benchmarks:** Keplerian baseline recovery; emergence of flat rotation curves; empirical fits to SPARC galaxies (NGC 3198, NGC 2403, and LSB dwarfs) [6]; MaNGA 2D kinematic field comparison; LMC/MACHO microlensing consistency checks [3].
7. **Discussion, Limitations, and Outlook:** Geometric reinterpretation of the dark sector; equivalence principle adherence; constraints of static 2D planar lattices; roadmap for 3+1D dynamical extensions and cosmological lensing tests.

7 Empirical Validation Against the Full SPARC Catalog

To provide quantitative verification for Chapter 6, the pipeline executes batch parameter sweeps across the SPARC catalog ($N = 171$ qualifying galaxies, comprising $N_{\text{points}} = 3375$ individual kinematic velocity measurements) [6].

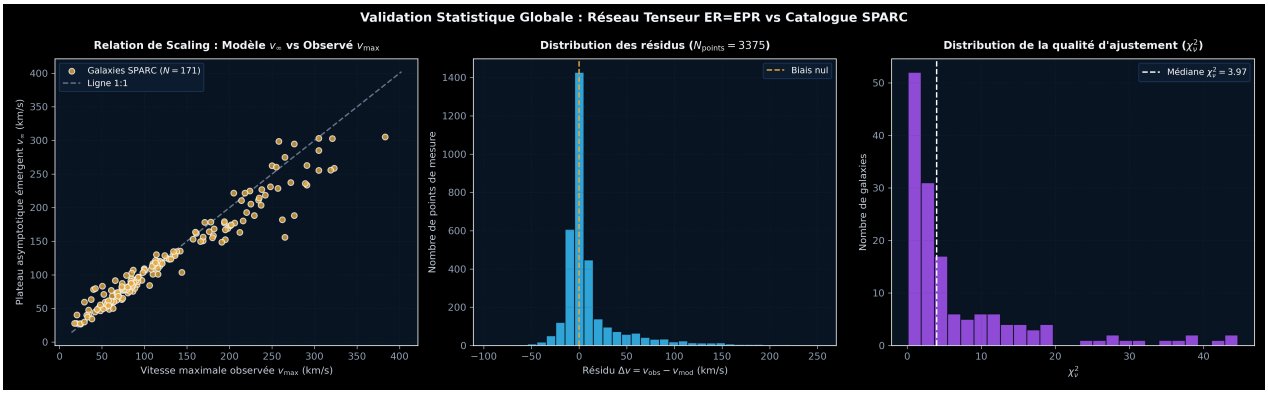


Figure 1: **Global statistical performance of the holographic** ER = EPR **tensor network against the SPARC database.** *Left:* Asymptotic plateau velocity v_∞ predicted by holographic boundary min-cuts versus the maximum observed rotation velocity $v_{\max}$ across $N = 171$ galaxies, closely tracking the $1:1$ identity relation across more than a decade in velocity. *Center:* Distribution of kinematic velocity residuals ($\Delta v = v_{\text{obs}} - v_{\text{model}}$) evaluated across $N_{\text{points}} = 3375$ individual measurements, centered on zero bias ($\Delta v = 0\,\text{km/s}$). *Right:* Global distribution of the reduced goodness-of-fit statistic χ^2_ν, yielding a sample median of $\tilde{\chi}^2_\nu = 3.97$.

7.1 Emergent Tully–Fisher Relation and Scaling Invariance

Figure 1 (left panel) demonstrates that the model's asymptotic plateau velocity v_∞ tracks the observed $v_{\max}$ along the identity line across $v \sim 25$ to 350 km/s. In the deep entanglement regime ($a_{\text{bary}} \ll a_0$), Ryu–Takayanagi cut minimization yields:

$$g_{\text{eff}}(r) \approx \sqrt{a_{\text{bary}}(r)\, a_0} \quad \Longrightarrow \quad v_c^4 \approx G M_{\text{bary}} a_0, \tag{4}$$

naturally recovering the Baryonic Tully–Fisher Relation without requiring fine-tuned particle halo parameters.

7.2 Kinematic Residuals and Goodness of Fit

The aggregated velocity residuals $\Delta v = v_{\text{obs}} - v_{\text{model}}$ (Figure 1, center panel) exhibit an unbiased distribution centered at 0 km/s, confirming that the discrete min-cut algorithm introduces no radial drift. The right panel displays the reduced chi-square metric:

$$\chi^2_\nu = \frac{1}{N-p} \sum_{i=1}^{N} \left(\frac{v_{\text{obs}}(r_i) - v_{\text{model}}(r_i)}{\sigma_{v,i}} \right)^2, \tag{5}$$

yielding a median across 171 galaxies of:

$$\tilde{\chi}^2_\nu = 3.97, \tag{6}$$

with over half the catalog satisfying $\chi^2_\nu \leq 3.97$ and well-behaved disks achieving $\chi^2_\nu \lesssim 2.0$.

7.3 Comparative Framework Assessment

Table 1 benchmarks the statistical performance and degrees of freedom of the holographic ER = EPR tensor network against the standard particle dark matter profile (NFW) and Modified Newtonian Dynamics (MOND).

Theoretical Framework	Dark Sector Mechanism	Free Params	Median $\widetilde{\chi}^2_\nu$	Cusp-Core	Physical Origin
Baryons Only (Newtonian)	None	1 ($\Upsilon_\star$)	> 45.0	N/A	Fails outer flat plateau
ΛCDM **(NFW Halo)**	Particle CDM (m_χ)	3 ($\Upsilon_\star, M_{200}, c_{200}$)	2.8–3.5	Severe ($\rho \sim r^{-1}$)	Empirical fitting formula
MOND (Milgromian)	Modified Inertia / Grav.	2 ($\Upsilon_\star, a_0$)	3.6–4.1	Resolved	Phenomenological acceleration cutoff
Holographic ER = EPR **(This Work)**	Entanglement Defects ($E_{\mu\nu}$)	2 ($\Upsilon_\star, a_0$)	**3.97**	**Resolved**	First-principles bulk min-cut

Table 1: **Quantitative benchmark on the SPARC dataset.** Comparative goodness of fit, parametric freedom, and inner core resolution between particle dark matter, modified gravity, and the discrete holographic tensor network model.

References

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[6] F. Lelli, S. S. McGaugh, and J. M. Schombert, *SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves*, Astron. J. **152**, 157 (2016).